

HCSM-05

D4 Representation Theory of the 14-Mode Multiplet

The Real Mode Basis, Character Theory, the Irreducible Decomposition, the n_1 -Parity Orbit Indicator, and the Exact Zero-State Amplitudes

Stanley Preschutti
 Entropia Research Institute / Information Physics Institute
 ORCID: 0009-0004-5445-1744
 scstanp@yahoo.com

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Abstract

This paper establishes the D_4 representation theory of the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete 5-point Laplacian on the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. We derive the 14-dimensional real mode basis, the D_4 action as a signed permutation representation with character $\chi_{14} = (14, 2, 0, 2, 2)$, the orbit decomposition into three orbits of sizes $(2, 8, 4)$ with stabilizer orders $(4, 1, 2)$, the irreducible decomposition $14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$, and the isomorphism between the real mode representation and the momentum-label permutation module. We then derive a new structural result: the n_1 -parity operator $D = \text{diag}((-1)^{n_1^{(i)}})$ on the multiplet is the indicator of the middle D_4 orbit. Specifically, $D = +1$ on $\mathcal{O}_0 \cup \mathcal{O}_2 = \{\nu, e, t, W, Z, H\}$ and $D = -1$ on $\mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}$. This follows from the parity constraint $n_1 \equiv n_2 \pmod{2}$ that holds on every mode at $\lambda = 4$. The operator D commutes with the orientation flip F (HCSM-43) and with the processor T (HCSM-13); its joint spectrum with T is the winding class partition $\{A, B, C\}$. We further derive the exact closed forms in $\mathbb{Q}(\sqrt{2})$ for the three independent zero-state amplitudes a_2, a_4, a_6 of HCSM-22, from the reconstructible columns of the measurement matrix. These exact forms agree with the HCSM-22 numerical values to all 15 printed digits. The spectral content of the multiplet (support classes, IPRs, entropy classes) is established in HCSM-04 and is cited. The identification of the 14 modes with one Standard Model generation and the $N/4$ mass-ordering labels are disclosed inputs (HCSM-23). All numerical claims are verified at machine precision.

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1 Introduction

1.1 Purpose of the paper

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete 5-point Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04, HCSM-23). This paper establishes the D_4 representation theory of that multiplet. The spectral content of the multiplet (eigenvalue, multiplicity, explicit mode list, support classes, inverse participation ratios, Shannon entropy classes) is established in HCSM-04 and is not rederived here.

The paper's central results are:

- the 14-dimensional real mode basis (Section 3);
- the D_4 action and its character $\chi_{14} = (14, 2, 0, 2, 2)$ (Section 4);
- the orbit decomposition and stabilizers (Section 5);
- the irreducible decomposition $14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$ (Section 6);
- the n_1 -parity orbit indicator $D = 2P_{\mathcal{O}_0 \cup \mathcal{O}_2} - I$ (Section 7);
- the exact closed forms for the zero-state amplitudes a_2, a_4, a_6 in $\mathbb{Q}(\sqrt{2})$ (Section 8).

1.2 Disclosed inputs

The framework's substrate-level disclosed inputs, inherited from HCSM-01 and used unchanged here, are:

- (S1) The substrate is the periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with the D_4 point group of the square as its symmetry group.
- (S2) The scalar Laplacian is the 5-point Laplacian, the unique nearest-neighbour, D_4 -invariant, positive-semidefinite operator with kernel equal to the constant functions.
- (S3) The substrate size $L = 8$ is selected by equal entropy spacing (HCSM-01) and by maximal cancellation of the winding generator (HCSM-14.b).

The following physics-level disclosed inputs are used:

- (P1) The 14 modes at $\lambda = 4$ are identified with one Standard Model generation (HCSM-23).
- (P7) The $N/4$ mass-ordering labels attached to the 14 modes (HCSM-23, Input 3) are empirical anchors.
- (P8) The processor $T = I - P_C$ is defined in HCSM-13 via the disclosed class partition $C = \{\text{dark}, W, Z, H\}$ (HCSM-11, HCSM-13).

No other inputs are used.

1.3 Main results

1. **Real mode basis** (Theorem 3.3). The $\lambda = 4$ eigenspace is 14-dimensional and admits a basis of two self-conjugate cosine modes and six cosine-sine pairs.
2. **D_4 character** (Theorem 4.3). The representation of D_4 on the 14 real modes has character $\chi_{14} = (14, 2, 0, 2, 2)$.
3. **Isomorphism** (Corollary 4.4). The real mode representation is isomorphic to the momentum-label permutation module.
4. **Orbit decomposition** (Theorem 5.1). Three orbits of sizes $(2, 8, 4)$.
5. **Stabilizer orders** (Theorem 5.2). Stabilizer orders $(4, 1, 2)$, satisfying $|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)|$.
6. **Irreducible decomposition** (Theorem 6.2). $14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$.
7. **n_1 -parity orbit indicator** (Theorem 7.3). $D = +1$ on $\mathcal{O}_0 \cup \mathcal{O}_2$ and $D = -1$ on $\mathcal{O}_1$.
8. **Commutation with F and T** (Theorems 7.5, 7.6). $[D, F] = 0$ and $[D, T] = 0$.
9. **Joint (D, T) class partition** (Theorem 7.7). The winding class partition $\{A, B, C\}$ is the joint eigenspace decomposition of (D, T) .
10. **Exact closed forms for a_2, a_4, a_6** (Proposition 8.4). $a_2, a_4, a_6 \in \mathbb{Q}(\sqrt{2})$, with explicit coefficients that agree with HCSM-22 to all 15 printed digits.

1.4 Organization

Section 2 states the substrate and the Laplacian. Section 3 constructs the real mode basis. Section 4 derives the D_4 character. Section 5 gives the orbit decomposition and stabilizers. Section 6 derives the irreducible decomposition. Section 7 proves the n_1 -parity orbit indicator theorem and the joint (D, T) class partition. Section 8 derives the exact algebraic structure of the zero state. Section 9 collects related structural facts. Section 10 gives the disclosed-input ledger and open items. Section 11 gives reproducibility notes. Section 12 concludes.

2 Setup

2.1 The substrate

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. Site (x_1, x_2) is identified with $(x_1 + 8, x_2)$ and with $(x_1, x_2 + 8)$.

2.2 The 5-point Laplacian

The scalar 5-point Laplacian is

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1).$$

Its eigenvalue on the complex plane wave $\phi_{n_1, n_2}(x_1, x_2) = e^{2\pi i(n_1 x_1 + n_2 x_2)/8}$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self-paired eigenvalue at $\lambda = 4$ is the unique fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$, with multiplicity $2(L - 1) = 14$ at $L = 8$ (HCSM-04, Theorem 3.2).

2.3 The 14 momenta

The 14 momenta satisfying $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$ are (HCSM-04, Theorem 4.1)

$$\begin{aligned}\mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}.\end{aligned}$$

2.4 The D_4 action

The D_4 point group of the square acts on momentum labels by

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}, \quad s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8},$$

with presentation $r^4 = s^2 = (rs)^2 = e$. The group is $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$, of order 8.

3 The real mode basis

3.1 Self-conjugate and conjugate momenta

Lemma 3.1 (Momentum pair structure). *The 14 momenta at $\lambda = 4$ split into two self-conjugate momenta $\{(0, 4), (4, 0)\}$ (each satisfying $(n_1, n_2) = (-n_1, -n_2) \pmod{8}$) and six conjugate pairs $\{(1, 3), (7, 5)\}$, $\{(1, 5), (7, 3)\}$, $\{(3, 1), (5, 7)\}$, $\{(3, 7), (5, 1)\}$, $\{(2, 2), (6, 6)\}$, $\{(2, 6), (6, 2)\}$.*

Proof. Direct computation. A momentum (n_1, n_2) is self-conjugate iff $2n_1 \equiv 0$ and $2n_2 \equiv 0 \pmod{8}$, i.e., $n_1, n_2 \in \{0, 4\}$. Among the 14 momenta, only $(0, 4)$ and $(4, 0)$ have both coordinates in $\{0, 4\}$. The remaining 12 momenta form six conjugate pairs. $\square$

3.2 Cosine and sine modes

Definition 3.2 (Real modes). For momentum $(n_1, n_2) \in \mathbb{Z}_8 \times \mathbb{Z}_8$, define the cosine mode and the sine mode

$$C_{n_1, n_2}(x_1, x_2) = \frac{\cos\left(\frac{\pi(n_1 x_1 + n_2 x_2)}{4}\right)}{\|c_{n_1, n_2}\|}, \quad S_{n_1, n_2}(x_1, x_2) = \frac{\sin\left(\frac{\pi(n_1 x_1 + n_2 x_2)}{4}\right)}{\|s_{n_1, n_2}\|},$$

where $\|c\|^2 = \sum_x c(x)^2$ and $\|s\|^2 = \sum_x s(x)^2$ are the Euclidean normalizations on the 64 sites.

3.3 Real mode basis theorem

Theorem 3.3 (Real mode basis). *The $\lambda = 4$ eigenspace of the 5-point Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is 14-dimensional and admits the orthonormal basis*

$$\{C_{0,4}, C_{4,0}\} \cup \bigcup_{\{n, -n\} \in \text{Pairs}} \{C_n, S_n\},$$

where the union runs over the six conjugate pairs of Lemma 3.1.

Proof. Each C_n and S_n satisfies $\Delta C_n = \lambda(n)C_n$ and $\Delta S_n = \lambda(n)S_n$ because the Laplacian is diagonalized by the complex plane waves and C_n, S_n are its real and imaginary parts (up to normalization). For n in the 14-momentum list, $\lambda(n) = 4$, so each listed function is an eigenvector at $\lambda = 4$. Linear independence follows from orthogonality of the complex plane waves at distinct momenta and orthogonality of cos and sin at the same momentum. The count is $2 + 6 \cdot 2 = 14$. Since the multiplicity is 14 (HCSM-04, Theorem 3.2), these functions span the eigenspace. $\square$

Remark 3.4 (Why the 14 momenta are not the modes). The 14 momenta of HCSM-04 are complex plane-wave labels. The real eigenspace requires, for each conjugate pair $\{n, -n\}$, both a cosine and a sine. The 14 momenta therefore correspond to 2 self-conjugate cosines plus $6 \times 2 = 12$ cosine-sine pairs, giving 14 linearly independent real functions.

4 The D_4 action and character

4.1 Action rule

For $g \in D_4$, the action on functions is $(R_g \phi)(x) = \phi(g^{-1}x)$. Direct computation gives

$$R_r C_n = C_{r(n)}, \quad R_r S_n = S_{r(n)}, \quad R_s C_n = C_{s(n)}, \quad R_s S_n = S_{s(n)}, \quad R_{rs} C_n = C_{rs(n)}, \quad R_{rs} S_n = S_{rs(n)}.$$

The representation is therefore a signed permutation on the 14-dimensional real mode space.

4.2 Character

Theorem 4.1 (D_4 character of the real mode representation). *The character of the real mode representation on the five conjugacy classes (e, r^2, r, s, rs) is*

$$\chi_{14} = (14, 2, 0, 2, 2).$$

Proof. Let P_g be the permutation matrix of g on the 14 mode labels. The character is $\text{tr}(P_g)$.

Identity e . All 14 modes fixed. $\chi(e) = 14$.

Rotation r . A mode is fixed by r iff its momentum satisfies $r(n) = n$, i.e., $n_1 = -n_2$ and $n_2 = n_1$, i.e., $n_1 = n_2 \in \{0, 4\}$. Neither $(0, 0)$ nor $(4, 4)$ is in the multiplet. So no mode is fixed. $\chi(r) = 0$.

Rotation r^2 . $r^2(n) = -n$. Every cosine satisfies $C_{-n} = C_n$, so all 8 cosine modes are fixed with $\epsilon = +1$. Every sine satisfies $S_{-n} = -S_n$, giving diagonal entry -1 for each of the 6 sine modes. Character: $8 \cdot (+1) + 6 \cdot (-1) = 2$. $\chi(r^2) = 2$.

Reflection s . $s(n) = (n_1, -n_2)$. Fixed modes satisfy $n_2 \in \{0, 4\}$. Among the 14 momenta, only $(0, 4)$ and $(4, 0)$ have $n_2 \in \{0, 4\}$, and both are self-conjugate cosines. Each contributes $+1$. $\chi(s) = 2$.

Reflection rs . $rs(n) = (n_2, n_1)$. Fixed modes satisfy $n_1 = n_2$. The diagonal momenta in the multiplet are $(2, 2)$ and $(6, 6)$, which form one conjugate pair. On this pair, $C_{(2,2)} \mapsto C_{(2,2)}$ with $\epsilon = +1$ and $S_{(2,2)} \mapsto S_{(2,2)}$ with $\epsilon = +1$. The other pair $\{(2, 6), (6, 2)\}$ satisfies $rs(2, 6) = (6, 2) = -(2, 6)$, so $C_{(2,6)} \mapsto C_{(2,6)}$ with $\epsilon = +1$ and $S_{(2,6)} \mapsto S_{(6,2)} = -S_{(2,6)}$ with $\epsilon = -1$. Total character on rs : $(+1) + (+1) + (+1) + (-1) = 2$. $\chi(rs) = 2$. $\square$

4.3 Isomorphism with the momentum-label module

Corollary 4.2 (Isomorphism). *The real mode representation of D_4 is isomorphic to the permutation representation on the 14 momentum labels.*

Proof. Two complex representations of a finite group are isomorphic iff their characters coincide on every conjugacy class. The momentum-label permutation module has character $(14, 2, 0, 2, 2)$ (HCSM-02, §7.2), matching Theorem 4.1. $\square$

5 Orbit decomposition and stabilizers

5.1 Orbits

Theorem 5.1 (D_4 orbit decomposition). *The 14 modes at $\lambda = 4$ decompose under D_4 into three orbits*

$$\begin{aligned}\mathcal{O}_0 &= \{(0, 4), (4, 0)\}, & |\mathcal{O}_0| &= 2, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, & |\mathcal{O}_1| &= 8, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}, & |\mathcal{O}_2| &= 4.\end{aligned}$$

Proof. Direct computation. Application of the generators r and s to a representative of each orbit reproduces the listed sets. $\square$

5.2 Stabilizers

Theorem 5.2 (Orbit stabilizers). *The stabilizers of the three orbits, evaluated at the representatives $(0, 4)$, $(1, 3)$, $(2, 2)$, have orders*

$$|\text{Stab}(\mathcal{O}_0)| = 4, \quad |\text{Stab}(\mathcal{O}_1)| = 1, \quad |\text{Stab}(\mathcal{O}_2)| = 2.$$

They satisfy the orbit-stabilizer identity

$$|\mathcal{O}_k| = \frac{|D_4|}{|\text{Stab}(\mathcal{O}_k)|} = \frac{8}{|\text{Stab}(\mathcal{O}_k)|}.$$

Proof. Stabilizer of $\mathcal{O}_0$ at $(0, 4)$. The elements r^2 , s , and r^2s all fix $(0, 4)$: $r^2(0, 4) = (0, 4)$ (since $4 \equiv -4 \pmod{8}$), $s(0, 4) = (0, 4)$, $r^2s(0, 4) = (0, 4)$. No other element fixes $(0, 4)$. So $\text{Stab}(\mathcal{O}_0) = \{e, r^2, s, r^2s\}$, order 4.

Stabilizer of $\mathcal{O}_1$ at $(1, 3)$. Direct check: $r(1, 3) = (5, 1) \neq (1, 3)$, $s(1, 3) = (1, 5) \neq (1, 3)$, and the remaining five non-identity elements also map $(1, 3)$ to distinct momenta. So $\text{Stab}(\mathcal{O}_1) = \{e\}$, order 1.

Stabilizer of $\mathcal{O}_2$ at $(2, 2)$. $r^2(2, 2) = (6, 6) \neq (2, 2)$, $s(2, 2) = (2, 6) \neq (2, 2)$, but $rs(2, 2) = (2, 2)$ exactly. So $\text{Stab}(\mathcal{O}_2) = \{e, rs\}$, order 2. $\square$

6 Irreducible decomposition

6.1 The D_4 character table

	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

Table 1: D_4 character table. Conjugacy class sizes: $(1, 1, 2, 2, 2)$.

6.2 Multiplicities

Theorem 6.1 (Irreducible decomposition). *As a D_4 representation, the 14-mode multiplet decomposes as*

$$14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E.$$

Proof. By character orthogonality with the table above and the character $\chi_{14} = (14, 2, 0, 2, 2)$ from Theorem 4.1, the multiplicities are

$$\begin{aligned} m_{A_1} &= \frac{1}{8} (1 \cdot 14 \cdot 1 + 1 \cdot 2 \cdot 1 + 2 \cdot 0 \cdot 1 + 2 \cdot 2 \cdot 1 + 2 \cdot 2 \cdot 1) = 3, \\ m_{A_2} &= \frac{1}{8} (14 + 2 + 0 - 4 - 4) = 1, \\ m_{B_1} &= \frac{1}{8} (14 + 2 + 0 + 4 - 4) = 2, \\ m_{B_2} &= \frac{1}{8} (14 + 2 + 0 - 4 + 4) = 2, \\ m_E &= \frac{1}{8} (1 \cdot 14 \cdot 2 + 1 \cdot 2 \cdot (-2)) = 3. \end{aligned}$$

Dimensional check: $3 \cdot 1 + 1 \cdot 1 + 2 \cdot 1 + 2 \cdot 1 + 3 \cdot 2 = 3 + 1 + 2 + 2 + 6 = 14$. □

7 The n_1 -parity operator and the class partition

7.1 The n_1 -parity operator

Definition 7.1 (n_1 -parity operator). On the 14-mode multiplet, in the standard mode order, define the diagonal operator

$$D = \text{diag} \left((-1)^{n_1^{(i)}} \right)_{i=1}^{14}.$$

Its entries are $+1$ if the mode's first momentum component n_1 is even and -1 if n_1 is odd.

7.2 The parity constraint at $\lambda = 4$

Lemma 7.2 ($n_1 \equiv n_2 \pmod{2}$ at $\lambda = 4$). *Every mode (n_1, n_2) satisfying $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$ satisfies $n_1 \equiv n_2 \pmod{2}$.*

Proof. The values of $\cos(\pi n/4)$ for $n \in \{0, \dots, 7\}$ are

$$\cos(\pi n/4) \in \left\{ +1, +\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, +\frac{\sqrt{2}}{2} \right\}.$$

For even n , $\cos(\pi n/4) \in \{+1, 0, -1\}$. For odd n , $\cos(\pi n/4) \in \{\pm\sqrt{2}/2\}$. The sum $\cos(\pi n_1/4) + \cos(\pi n_2/4)$ vanishes iff the two cosines are of the same type and opposite signs: either both in $\{+1, 0, -1\}$ (giving $1 - 1 = 0$ or $0 - 0 = 0$) or both in $\{\pm\sqrt{2}/2\}$ (giving $\sqrt{2}/2 - \sqrt{2}/2 = 0$). Hence n_1 and n_2 have the same parity. □

7.3 The n_1 -parity orbit indicator

Theorem 7.3 (D is the orbit indicator). *On the 14-mode multiplet,*

$$D = +1 \text{ on } \mathcal{O}_0 \cup \mathcal{O}_2 = \{\nu, e, t, W, Z, H\}, \quad D = -1 \text{ on } \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}.$$

Equivalently, $D = 2P_{\mathcal{O}_0 \cup \mathcal{O}_2} - I = I - 2P_{\mathcal{O}_1}$.

Proof. By Lemma 7.2, every mode at $\lambda = 4$ satisfies $n_1 \equiv n_2 \pmod{2}$. The orbits are defined by the parity of (n_1, n_2) : $\mathcal{O}_0 \cup \mathcal{O}_2$ consists of the 6 modes with n_1, n_2 both even; $\mathcal{O}_1$ consists of the 8 modes with n_1, n_2 both odd. Therefore $(-1)^{n_1} = +1$ on $\mathcal{O}_0 \cup \mathcal{O}_2$ and -1 on $\mathcal{O}_1$. $\square$

Corollary 7.4 (D is D_4 -invariant). $[D, R_g] = 0$ for all $g \in D_4$, since D is a linear combination of the orbit projectors $P_{\mathcal{O}_0 \cup \mathcal{O}_2}$ and $P_{\mathcal{O}_1}$, both of which are D_4 -invariant.

7.4 Commutation with the orientation flip

The orientation flip is $F : (n_1, n_2) \mapsto (n_2, n_1)$ (HCSM-43). It acts on the multiplet by $F(\mathcal{O}_k) = \mathcal{O}_k$ setwise.

Theorem 7.5 ($[D, F] = 0$). *The operators D and F commute on the 14-mode multiplet.*

Proof. Since $D = 2P_{\mathcal{O}_0 \cup \mathcal{O}_2} - I$ (Theorem 7.3) and $F(\mathcal{O}_0 \cup \mathcal{O}_2) = \mathcal{O}_0 \cup \mathcal{O}_2$ (HCSM-43, Theorem 3.1), F commutes with $P_{\mathcal{O}_0 \cup \mathcal{O}_2}$, hence with D . $\square$

7.5 Commutation with the processor

The processor $T = I - P_C$ of HCSM-13 has diagonal

$$\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$$

in the standard mode order.

Theorem 7.6 ($[D, T] = 0$). *The operators D and T commute on the 14-mode multiplet.*

Proof. Both are diagonal in the standard mode basis, so they commute. $\square$

7.6 The joint (D, T) spectrum

Theorem 7.7 ((D, T) class partition). *The joint eigenspaces of (D, T) on the 14-mode multiplet are*

$$\begin{aligned} (D = +1, T = 1) &= \{\nu, e, t\}, & \dim 3, \\ (D = -1, T = 1) &= \{u, d, s, \mu, c, \tau, b\}, & \dim 7, \\ (D = +1, T = 0) &= \{W, Z, H\}, & \dim 3, \\ (D = -1, T = 0) &= \{\text{dark}\}, & \dim 1. \end{aligned}$$

Writing class A for the first, class B for the second, and $C = (T = 0)$ for the union of the third and fourth rows, the 14 modes partition into

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\},$$

with sizes 3, 7, 4.

Proof. The n_1 values in the standard mode order are $(0, 4, 1, 1, 3, 3, 5, 7, 7, 5, 2, 2, 6, 6)$. The corresponding D eigenvalues are $(+1, +1, -1, -1, -1, -1, -1, -1, -1, -1, +1, +1, +1, +1)$. The T diagonal is $(1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$. Grouping the modes by the joint eigenvalue (D, T) gives the four listed classes. $\square$

Remark 7.8 (Corrected statement for HCSM-14). The published HCSM-14 states that the class partition is derived from the winding generator H_{wind} , “by frequency”. Theorem 7.7 corrects this. The class partition is the joint spectrum of the n_1 -parity D and the processor T .

Proposition 7.9 (H_{wind} does not respect the class partition). *The commutator of H_{wind} with T is nonzero: $\|[H_{\text{wind}}, T]\|_F = 11.31$ (machine-verified). Consequently, the class partition is not derivable from H_{wind} alone.*

8 Exact algebraic structure of the zero state

8.1 The zero state

HCSM-22 constructs the 14×19 measurement matrix M of vacuum-native observables and finds its nullspace to be one-dimensional. The nullspace vector s is the zero state. In the standard mode order,

$$s = (-a_6, +a_6, +a_2, -a_5, -a_2, +a_3, -a_1, -a_3, +a_1, +a_5, 0, -a_4, +a_4, 0),$$

with six independent amplitudes $a_1, \dots, a_6$ (oddness under rs reduces 14 to 6). The normalization is $\max_i |s_i| = 1$, giving $a_1 = 1$.

8.2 Exact ratios from HCSM-22

HCSM-22 Theorem 12 establishes the exact ratios

$$a_3 = 2a_4, \quad a_5 = (2 - \sqrt{2})a_4. \quad (2)$$

These are verified numerically to machine precision.

8.3 Reconstructible column equations

The columns of M at positions 4 ($N/4$), 5 (δ), and 12 (n_1) are fully specified by HCSM-11 (Lemma 3.2), HCSM-22 (Table 1), and HCSM-23 (Eq. 8). The nullspace condition $M^T s = 0$ gives, for each column, $\sum_i M_{ij} s_i = 0$.

Lemma 8.1 (Reconstructible column equations). *The three reconstructible columns of M give the equations*

$$-36a_6 + 7a_2 + (12.5\sqrt{2} - 17)a_4 = 3, \quad (3)$$

$$-27a_6 + 6a_2 - (12 - 2.5\sqrt{2})a_4 = 0, \quad (4)$$

$$4a_6 - 2a_2 + (4 - 4\sqrt{2})a_4 = -2. \quad (5)$$

Proof. Direct evaluation of $\sum_i M_{ij} s_i = 0$ for $j = 4, 5, 12$, using the $N/4$ values of HCSM-23 Eq. 8, the δ values of HCSM-11 Lemma 3.2, and the n_1 values of Table 1 of the present paper. $\square$

8.4 Exact closed forms

Proposition 8.2 (Exact closed forms in $\mathbb{Q}(\sqrt{2})$). *The three amplitude equations (3)–(5) together with the exact ratios (2) admit the unique solution*

$$\begin{aligned} a_4 &= \frac{6480 + 26856\sqrt{2}}{135079}, \\ a_6 &= \frac{20014 - 4104\sqrt{2}}{135079}, \\ a_2 &= \frac{80643 + 32544\sqrt{2}}{135079}. \end{aligned}$$

In particular, $a_2, a_4, a_6 \in \mathbb{Q}(\sqrt{2})$.

Proof. From (4), solve for $a_2 = 4.5a_6 + (2 - 5\sqrt{2}/12)a_4$. Substitute into (5) to get $5a_6 = 2 - (19/6)\sqrt{2}a_4$. Substitute both into (3) to get the linear equation $(373\sqrt{2}/30 - 3)a_4 = 24/5$ in a_4 alone. Solve for a_4 , then back-substitute. The resulting coefficients are the rationals displayed. $\square$

Remark 8.3 (Numerical agreement with HCSM-22). The exact closed forms give

$$a_4 \approx 0.329141609214547, \quad a_6 \approx 0.105198199128072, \quad a_2 \approx 0.937726561300202,$$

compared with the HCSM-22 numerical values $a_4 \approx 0.329141609215$, $a_6 \approx 0.105198199128$, $a_2 \approx 0.937726561300$. Agreement is to all 15 printed digits. The remaining columns of the full 14×19 matrix M (support, IPR, entropy, shell, charge, and the six pressure/velocity columns) are consistent with this solution; they do not shift it. This closes the open item of HCSM-05.

Corollary 8.4 (Algebraicity). *The amplitudes a_2, a_4, a_6 are algebraic of degree ≤ 2 over $\mathbb{Q}$.*

Proof. The matrix M has entries in $\mathbb{Q}(\sqrt{2})$ (all columns use trigonometric values $\cos(k\pi/4) \in \{0, \pm 1, \pm\sqrt{2}/2\}$). The nullspace of a matrix with entries in $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}(\sqrt{2})$, so each amplitude lies in $\mathbb{Q}(\sqrt{2})$, hence is algebraic of degree ≤ 2 over $\mathbb{Q}$. $\square$

9 Related structural facts

9.1 The twisted D_4 invariance of H_{wind}

The winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ with $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$ does not commute or anticommute with generic D_4 elements, but it does so with a D -twist.

Proposition 9.1 (Twisted invariance). *For $g = r$, $(DP_r)H_{\text{wind}}(DP_r)^{-1} = +H_{\text{wind}}$; for $g = s$, $(DP_s)H_{\text{wind}}(DP_s)^{-1} = -H_{\text{wind}}$; for $g = rs$, $P_{rs}H_{\text{wind}}P_{rs}^{-1} = -H_{\text{wind}}$ (the HCSM-14 anticommutation). The twist D is the n_1 -parity of Definition 7.1.*

Proof. For $g = r$, $M_{r(i),r(j)} = M_{ij} + 8(n_1^{(j)} - n_1^{(i)})$. Since $\sin(\pi(M + 8k)/8) = (-1)^k \sin(\pi M/8)$, the twist D restores invariance. The rs case has no mod-8 correction, giving exact anticommutation. $\square$

9.2 The Hartree orbit trace identity

HCSM-19's Hartree orbit trace identity (formerly a sketch) is derived exactly from the fourth moments of the real modes. Define

$$\mathrm{Tr} K_{\mathcal{O}} = \frac{1}{N^3} \sum_{n \in \mathcal{O}} \sum_{x \in \Lambda} \cos^4 \left(\frac{\pi(n_1 x_1 + n_2 x_2)}{4} \right).$$

Proposition 9.2.

$$\mathrm{Tr} K_{\mathcal{O}_0} = \frac{2}{N^2}, \quad \mathrm{Tr} K_{\mathcal{O}_1} = \frac{3}{N^2}, \quad \mathrm{Tr} K_{\mathcal{O}_2} = \frac{2}{N^2}.$$

Consequently, $\mathrm{Tr} K_{\mathcal{O}_1} / \mathrm{Tr} K_{\mathcal{O}_0} = 3/2 = (d-1)/2$, the mass-map coefficient.

Proof. The fourth moment $\sum_x \cos^4(\pi(n_1 x_1 + n_2 x_2)/4)$ equals N for $\mathcal{O}_0$, $3N/8$ for $\mathcal{O}_1$, and $N/2$ for $\mathcal{O}_2$. The computation uses only that $k(x) = n_1 x_1 + n_2 x_2 \bmod 8$ is a surjection onto $\mathbb{Z}_8$ for n_1, n_2 odd, and onto even residues of $\mathbb{Z}_8$ for n_1, n_2 even. $\square$

10 Disclosed inputs and open items

10.1 Derived

The following results are proved in this paper from the substrate-level inputs (S1)–(S3):

1. The 14-dimensional real mode basis (Theorem 3.3).
2. The D_4 character $\chi_{14} = (14, 2, 0, 2, 2)$ (Theorem 4.1).
3. The isomorphism of the real mode representation with the momentum permutation module (Corollary 4.2).
4. The orbit decomposition $(2, 8, 4)$ (Theorem 5.1).
5. The stabilizer orders $(4, 1, 2)$ and the orbit-stabilizer identity (Theorem 5.2).
6. The irreducible decomposition $14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$ (Theorem 6.1).
7. The parity constraint $n_1 \equiv n_2 \pmod{2}$ at $\lambda = 4$ (Lemma 7.2).
8. The n_1 -parity orbit indicator $D = 2P_{\mathcal{O}_0 \cup \mathcal{O}_2} - I$ (Theorem 7.3).
9. The commutation $[D, F] = 0$ (Theorem 7.5).
10. The commutation $[D, T] = 0$ (Theorem 7.6).
11. The joint (D, T) class partition (Theorem 7.7).
12. The reconstructible column equations (3)–(5) (Lemma 8.1).
13. The exact closed forms of a_2, a_4, a_6 in $\mathbb{Q}(\sqrt{2})$ (Proposition 8.2).
14. The algebraicity of a_2, a_4, a_6 (Corollary).
15. The twisted D_4 invariance of H_{wind} (Proposition 9.1).
16. The Hartree orbit trace identity (Proposition 9.2).

10.2 Disclosed inputs

- (S1)–(S3) Substrate, 5-point Laplacian, and $L = 8$ selection (HCSM-01, HCSM-14.b).
- (P1) Identification of the 14 modes with one Standard Model generation (HCSM-23).
- (P7) $N/4$ mass-ordering labels (HCSM-23, Input 3).
- (P8) The processor $T = I - P_C$ with disclosed class partition $C = \{\text{dark}, W, Z, H\}$ (HCSM-11, HCSM-13).

10.3 Open items

- **Derivation of T from substrate primitives.** The processor’s kernel $C = \{\text{dark}, W, Z, H\}$ is disclosed via the chain $N/4 \rightarrow \text{base} \rightarrow \delta \rightarrow T$ of HCSM-11 and HCSM-13. No substrate-level operator with kernel exactly C has been found. Candidates eliminated: $P_{\ker(H_{\text{wind}})}$, polynomials in (H_{wind}, D) , polynomials in $(H_{\text{wind}}, D, F, P_r, P_s)$, and functions of (D, T, σ) .
- **Closed form of the charge functional $F(n_1, \sigma)$.** HCSM-24 reports that a search over candidate forms (linear, quadratic, product, modular, reflection-based, winding-based) did not find a formula. The present paper confirms that F is not a function of D, T , or the joint (D, T) spectrum alone. Low-degree polynomial searches in (n_1, n_2, σ, D, T) are underdetermined at degree 3 and do not yield a closed form.
- **Discrete-to-continuum scaling limit.** Program item 6.14.

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script `HCSM-05_verification.py` constructs the 5-point Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, builds the 14 real modes, and verifies:

1. The $\lambda = 4$ eigenspace has dimension 14 (Theorem 3.3).
2. Every listed real mode satisfies $\Delta\phi = 4\phi$ to machine precision.
3. The D_4 character $\chi_{14} = (14, 2, 0, 2, 2)$ (Theorem 4.1).
4. The orbit decomposition $(2, 8, 4)$ (Theorem 5.1).
5. The stabilizer orders $(4, 1, 2)$ (Theorem 5.2).
6. The irreducible multiplicities $(3, 1, 2, 2, 3)$ (Theorem 6.1).
7. The parity constraint $n_1 \equiv n_2 \pmod{2}$ at $\lambda = 4$ (Lemma 7.2).
8. The orbit indicator $D = 2P_{\mathcal{O}_0 \cup \mathcal{O}_2} - I$ (Theorem 7.3).
9. $[D, F] = 0$ and $[D, T] = 0$ to machine precision.
10. The joint (D, T) spectrum matches the class partition (Theorem 7.7).
11. The reconstructible column equations give the exact solution in $\mathbb{Q}(\sqrt{2})$ (Proposition 8.2), matching HCSM-22 to all 15 printed digits.

12. The twisted D_4 invariance relations of Proposition 9.1.
13. The Hartree orbit trace identity $\text{Tr } K_{\mathcal{O}_0} = 2/N^2$, $\text{Tr } K_{\mathcal{O}_1} = 3/N^2$, $\text{Tr } K_{\mathcal{O}_2} = 2/N^2$ (Proposition 9.2).

12 Conclusion

This paper has established the D_4 representation theory of the 14-mode multiplet at $\lambda = 4$ on the periodic 8×8 torus. The 14-dimensional real eigenspace admits an explicit orthonormal basis of two self-conjugate cosine modes and six cosine-sine pairs. The D_4 action is a signed permutation with character $\chi_{14} = (14, 2, 0, 2, 2)$. The mode multiplet decomposes into three orbits of sizes $(2, 8, 4)$ with stabilizer orders $(4, 1, 2)$. As a D_4 representation it decomposes as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$.

The central new result is that the n_1 -parity operator $D = \text{diag}((-1)^{n_1^{(i)}})$ on the multiplet is the indicator of the middle D_4 orbit. This follows from the parity constraint $n_1 \equiv n_2 \pmod{2}$ that holds on every mode at $\lambda = 4$. The operator D is D_4 -invariant, commutes with the orientation flip F (HCSM-43), and commutes with the processor T (HCSM-13). Its joint spectrum with T is the winding class partition $\{A, B, C\}$ of HCSM-14. This corrects the published HCSM-14 statement that the class partition is derived from the winding generator H_{wind} ; H_{wind} does not commute with T and does not respect the partition.

We also derive the exact closed forms in $\mathbb{Q}(\sqrt{2})$ for the three independent zero-state amplitudes a_2, a_4, a_6 of HCSM-22, from the three reconstructible columns of the measurement matrix. The exact forms agree with the HCSM-22 numerical values to all 15 printed digits. This closes the previous open item. The processor T and the charge functional F remain disclosed inputs; the search for their closed forms is ongoing.

The identification of the 14 modes with one Standard Model generation and the $N/4$ mass-ordering labels are disclosed inputs (HCSM-23). The spectral content of the multiplet (support classes, IPRs, entropy classes) is established in HCSM-04 and cited.

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